WORKED SOLUTION

STEP II 2015, Question 8

Pure • Vectors and Tangent Circles

STEP II 2015, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q8

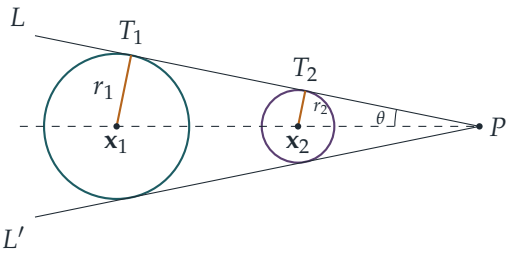
The key idea

Each centre is the same distance from both tangents (its own radius), so both centres lie on the line that bisects the angle between L and L'. Along that line, distance from P is proportional to radius, which gives the formula in (i) at once. For (ii), dividing the formula by $r_1 r_2$ makes it far more symmetric: in terms of $\mathbf{x}_i / r_i$ and $1/r_i$ the three foci satisfy one linear relation whose coefficients add up to zero, and that is exactly what collinearity means.

Solution

Part (i)

Let θ be half the angle between L and L' at P, in the angle that contains the circles.



A tangent to a circle is at perpendicular distance equal to the radius from the centre. So $\mathbf{x}_1$ is at distance r_1 from both L and L', and points equidistant from two intersecting lines lie on one of the two angle bisectors at P. Since both circles lie on the same side of L and on the same side of L', both centres are in the same one of the four angular regions cut out by L and L', so they lie on the *same* half-line from P: the bisector of that region. Let $\mathbf{u}$ be the unit vector along it, pointing from P towards the circles.

If T_1 is the point where L touches C_1, the triangle PT_1X_1 (with X_1 the centre) has a right

angle at T_1, so $r_1 = PX_1 \sin\theta$. In the same way $r_2 = PX_2 \sin\theta$. Hence

$$\mathbf{x}_1 - \mathbf{p} = \frac{r_1}{\sin\theta}\,\mathbf{u}, \qquad \mathbf{x}_2 - \mathbf{p} = \frac{r_2}{\sin\theta}\,\mathbf{u},$$

where $\mathbf{p}$ is the position vector of P. Eliminating $\mathbf{u}/\sin\theta$,

$$r_2(\mathbf{x}_1 - \mathbf{p}) = r_1(\mathbf{x}_2 - \mathbf{p}) \quad \Longrightarrow \quad (r_1 - r_2)\,\mathbf{p} = r_1\mathbf{x}_2 - r_2\mathbf{x}_1.$$

The circles have different sizes, so $r_1 \neq r_2$ and we may divide:

$$\mathbf{p} = \frac{r_1\mathbf{x}_2 - r_2\mathbf{x}_1}{r_1 - r_2},$$

as required. (Equal radii would make L and L' parallel, with no focus at all.)

Part (ii)

Ask yourself The formula for $\mathbf{p}$ is not symmetric in the obvious way. Can we rewrite it so the three foci look alike?

Divide $(r_1 - r_2)\mathbf{p} = r_1\mathbf{x}_2 - r_2\mathbf{x}_1$ by $r_1 r_2$. Writing $s_i = 1/r_i$ and $\mathbf{y}_i = \mathbf{x}_i/r_i$, it becomes

$$(s_2 - s_1)\,\mathbf{p} = \mathbf{y}_2 - \mathbf{y}_1.$$

Applying part (i) to the other two pairs of circles (allowed, as $r_3 \neq r_1$ and $r_3 \neq r_2$, and the circles do not overlap) gives, with $\mathbf{q}$ and $\mathbf{r}$ the position vectors of Q and R,

$$(s_3 - s_1)\,\mathbf{q} = \mathbf{y}_3 - \mathbf{y}_1, \qquad (s_3 - s_2)\,\mathbf{r} = \mathbf{y}_3 - \mathbf{y}_2.$$

The right-hand sides satisfy $(\mathbf{y}_2 - \mathbf{y}_1) + (\mathbf{y}_3 - \mathbf{y}_2) = \mathbf{y}_3 - \mathbf{y}_1$, so

$$(s_2 - s_1)\,\mathbf{p} + (s_3 - s_2)\,\mathbf{r} = (s_3 - s_1)\,\mathbf{q}.$$

Since $s_3 \neq s_1$, we can write

$$\mathbf{q} = \lambda\,\mathbf{p} + (1 - \lambda)\,\mathbf{r}, \qquad \text{where } \lambda = \frac{s_2 - s_1}{s_3 - s_1},$$

because the two coefficients $\frac{s_2 - s_1}{s_3 - s_1}$ and $\frac{s_3 - s_2}{s_3 - s_1}$ add up to 1. So $\mathbf{q} = \mathbf{r} + \lambda(\mathbf{p} - \mathbf{r})$, which is a point on the line through R and P. Hence P, Q and R are collinear. (If P and R happen to coincide, this gives $Q = P = R$ as well, and the statement holds trivially.)

Part (iii)

From part (ii), $\mathbf{q} = \lambda\mathbf{p} + (1 - \lambda)\mathbf{r}$. Assuming $P \neq R$, the point $\lambda\mathbf{p} + (1 - \lambda)\mathbf{r}$ is the midpoint $\frac{1}{2}(\mathbf{p} + \mathbf{r})$ exactly when $\lambda = \frac{1}{2}$ (different values of λ give different points of the line). So the condition is

$$\frac{s_2 - s_1}{s_3 - s_1} = \frac{1}{2} \iff 2s_2 = s_1 + s_3 \iff \frac{2}{r_2} = \frac{1}{r_1} + \frac{1}{r_3} \iff r_2 = \frac{2r_1 r_3}{r_1 + r_3}.$$

In words, r_2 must be the harmonic mean of r_1 and r_3. (In the degenerate case $P = R$, all three foci coincide and Q is trivially half-way, whatever the radii.)

Checks

- Part (i): putting $r_2 \to 0$ shrinks C_2 to a point, and the formula gives $\mathbf{p} = \mathbf{x}_2$: the tangents from a point pass through that point. Also $\mathbf{p}$ is $\mathbf{x}_2 + \frac{r_2}{r_1 - r_2}(\mathbf{x}_2 - \mathbf{x}_1)$, beyond the smaller circle when $r_2 < r_1$, as in the diagram.
- A numerical case for (iii): take centres $(0,0)$, $(10,1)$, $(3,8)$ with radii 1, $\frac{3}{2}$, 3, so that $\frac{2}{3/2} = \frac{4}{3} = 1 + \frac{1}{3}$. Then $P = (-20,-2)$, $R = (17,-6)$ and $Q = (-\frac{3}{2},-4)$, which is indeed the midpoint of PR.
- The formulas were also confirmed by computer: intersecting the two tangent lines directly gives the same point P for many random pairs of circles.

Common slips

- Not explaining why P lies on the line of centres, or why P is outside the segment joining them (it is the *same* half-line from P that carries both centres).
- Showing that $\mathbf{q} - \mathbf{p}$ and $\mathbf{r} - \mathbf{p}$ are parallel without checking that the relation's coefficients sum to zero; a relation $\alpha\mathbf{p} + \beta\mathbf{q} + \gamma\mathbf{r} = \mathbf{0}$ only implies collinearity when $\alpha + \beta + \gamma = 0$.

Beyond STEP

The focus P is the centre of the enlargement (homothety) with positive scale factor r_2/r_1 that maps C_1 onto C_2. Part (ii) is Monge's three-circle theorem: composing the enlargements about Q and R gives the enlargement about P, and the centre of a composite of two enlargements lies on the line through their centres. The algebra in (ii) is a first taste of barycentric (affine) coordinates, where collinearity is a linear relation with coefficients summing to zero. Strang's *Introduction to Linear Algebra* develops the linear maps behind this, and Siklos's *Advanced Problems in Mathematics* has more questions of this kind.

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